Continuous versus Discrete Tactile Representations for Vision-Language-Action Avocado Sorting

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Abstract

Vision-language-action (VLA) policies typically have no direct channel for non-visual physical properties, such as tactile ripeness: such evidence is either dropped entirely or folded into the policy as a discrete signal, a classifier output substituted into a text prompt. The latter reuses the language channel VLA models are pretrained on, but compresses a spatially structured deformation field into a low-bandwidth categorical cue represented by only a few language tokens. We investigate how the pathway used to incorporate tactile evidence affects contact-rich manipulation. Using robotic avocado ripeness sorting as a testbed, where tactile deformation cues are inaccessible to vision, we compare three conditions: a vision-only π_0 baseline with no tactile signal, discrete text-prompt conditioning on a separately trained tactile classifier, and continuous fusion, where GelSight tactile images are encoded via a pretrained DINOv2 encoder and concatenated with visual embeddings prior to the π_0 backbone, without added cross-modal attention. In a small-scale evaluation on 10 held-out avocados unseen during demonstration collection, with each specimen tested twice per method, continuous fusion reaches a 95% sorting success rate, versus 80% for discrete text-prompt conditioning and 70% with no tactile signal, at only 7.7% inference overhead over the vision-only baseline; the discrete classifier matches the specimen label in all 20 of its robot trials. The central observation is that even a correct tactile category can fail to control a pretrained action policy when communicated only through a prompt.

Introduction

Automation is increasingly necessary in agri-food production, driven by the limitations of manual inspection and the inability of rule-based systems to handle natural variability in object appearance, geometry, and physical properties. Vision-language-action (VLA) models have emerged as a promising solution, offering strong generalization to such variability through large-scale pretraining on diverse manipulation demonstrations (Brohan et al. 2023; Kim et al. 2025; Black et al. 2024). However, a class of physical properties that determine food quality, such as surface compliance, firmness, and ripeness, remains inaccessible to cameras regardless of imaging conditions. Avocados exemplify this limitation: two avocados at different ripeness stages can appear near-identical under RGB imaging while exhibiting markedly different deformation patterns under contact.

Existing strategies for conditioning VLA policies on such properties propagate them to the action policy as discrete signals: classifier outputs, completion flags, or updated text prompts. These approaches share a common limitation. Quality-relevant physical properties are reduced to a low-dimensional categorical cue carried by a small number of language tokens, constraining the representational bandwidth available for conditioning manipulation on sensory feedback. This mismatch is not specific to food handling; any contact-rich task where task-relevant state is only observable through touch, such as assembly, cable routing, or fabric handling, faces the same bottleneck when tactile evidence is only permitted to enter the policy through the language channel. We frame this as a representation problem connecting to open questions on representations for robot learning and the choice of representation for non-visual sensory modalities. Our central claim is that a correct tactile category is not necessarily an effective control signal: when communicated only through a prompt, it can fail to steer a pretrained action policy. We test this claim through a robot evaluation of two practical tactile-integration pipelines.

Figure 1 illustrates the underlying perceptual ambiguity: RGB frames of ripe and firm avocados are nearly indistinguishable, while tactile frames from the same contacts reveal clearly different deformation patterns.

We address this gap with a continuous tactile-visual fusion approach for ripeness-conditioned robotic sorting. Here, *continuous* denotes a continuous-valued tactile representation, not continuous-time feedback; the Continuous Tactile-Visual Policy section below specifies the observation cadence. Dual GelSight tactile sensors are encoded with a pretrained DINOv2 encoder, and the resulting embeddings are concatenated with visual tokens ahead of a π_0 VLA backbone, enabling joint reasoning over both modalities without additional cross-modal attention mechanisms. The main contributions of this work are as follows:

- We develop a lightweight tactile-visual robotic sorting system that augments a pretrained VLA backbone with continuous tactile embeddings without adding cross-modal attention machinery.
- We present preliminary evidence, from a small-scale

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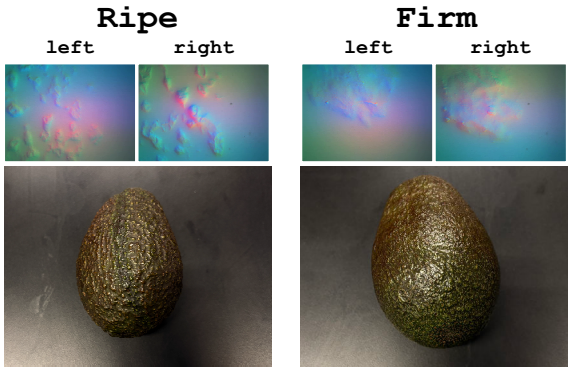


Figure 1: Ripe and firm avocados are visually near-identical under RGB imaging (bottom) but produce distinct tactile deformation patterns under contact (top): ripe avocados yield high-contrast, spatially complex patterns, while firm avocados yield smoother, more diffuse gradients.

robot evaluation against both a vision-only baseline and a discrete text-token baseline, that the tested continuous-fusion pipeline outperforms discrete conditioning even when the underlying tactile classifier reaches 100% validation accuracy.

- We characterize the three systems through held-out physical trials, class- and position-wise failures, uncertainty intervals, and inference-resource measurements.

Related Work

Tactile Sensing for Manipulation

Vision-based tactile sensors such as GelSight (Yuan, Dong, and Adelson 2017) capture high-resolution surface deformation during contact, revealing material properties invisible to cameras. Prior systems use these signals for material and texture discrimination (Yuan et al. 2018), grasp and regrasp control (Calandra et al. 2018), grasp-stability estimation (Kolamuri et al. 2021), and slip detection for fragile produce (Liu et al. 2024). Recent fusion methods condition intermediate visual features on tactile representations or activate tactile features selectively after contact (Morissette et al. 2026). Collectively, this literature shows that touch can improve contact-rich control, but most work evaluates motor outcomes such as insertion, stability, or slip recovery. Our task differs in that touch must first resolve a latent material property—ripeness—and that property must then select a long-horizon sorting destination.

Multimodal Food-Quality Assessment

A recent survey identifies tactile sensing as particularly important when visually similar produce items cannot be distinguished by appearance alone (Mandil et al. 2023). Prior multimodal quality-assessment work has combined accelerometer-based tactile sensing with near-infrared spectroscopy for mango grading (Cortés et al. 2017), used two-stage tactile-visual pipelines for avocado ripeness (Zhang et al. 2025b), and applied transformer architectures to tactile avocado grading (Fahmy et al. 2024). These studies establish that contact deformation carries useful ripeness information. However, they primarily optimize or evaluate a classifier separately from the downstream manipulation policy. A correct category prediction therefore does not by itself establish that a pretrained action model can reliably use the prediction to change its behavior. We study this interface explicitly by comparing a classifier-to-prompt pipeline with a policy that receives dense tactile features.

Vision-Language-Action Models and Touch

Pretrained vision-language backbones support VLA action generation. RT-2 established the approach of representing robot actions within a vision-language model (Brohan et al. 2023), OpenVLA extended open-source large-scale robot pretraining (Kim et al. 2025), and π_0 introduced flow matching to generate continuous action chunks (Black et al. 2024). Recent work has begun extending VLA models with tactile modalities for insertion and contact-rich manipulation (Zhang et al. 2025a), physical-knowledge generalization (Huang et al. 2025), and feature-wise tactile modulation (Morissette et al. 2026), alongside separate extensions for long-horizon task sequencing and weight-aware grasping (Yang et al. 2025; An et al. 2025). These tactile-VLA extensions largely target reactive motor control. Fusing quality-relevant tactile representations directly into the action policy, and comparing that pathway with discrete text conditioning while verifying the classifier output, remains comparatively underexplored.

Problem Formulation

Let $\mathcal{V} \subset \mathbb{R}^{H\times W\times 3}$ and $\mathcal{T} \subset \mathbb{R}^{H'\times W'\times 3}$ denote the RGB and vision-based tactile observation spaces. At action-chunk boundary t, the robot receives RGB observations $v_t \in \mathcal{V}$ from two cameras, tactile observations $\tau_t \in \mathcal{T}$ from two gripper-mounted sensors, a proprioceptive state $q_t \in \mathbb{R}^7$ containing six arm joints and one gripper coordinate, and tokenized task prompt p. A policy with parameters θ maps this multimodal context to an M-step action chunk,

$$A_t = f_\theta(v_t, q_t, p, \tau_t), \qquad A_t \in \mathbb{R}^{M\times 7}, \tag{1}$$

where $M = 50$ and every row specifies six joint-position targets and one gripper target.

The comparison concerns how τ_t reaches the policy. The discrete pipeline applies a tactile classifier c and maps its category to a prompt through a hand-designed function g, giving

$$A_t^{\text{disc}} = f_{\theta_d}(v_t, q_t, g(c(\tau_t))). \tag{2}$$

The continuous pipeline instead retains a sequence of patch features $\phi(\tau_t)$ and projects them into the VLA token dimension,

$$A_t^{\text{cont}} = f_{\theta_c}(v_t, q_t, p; W\phi(\tau_t)). \tag{3}$$

Here the semicolon indicates token-sequence augmentation, not a separate decision module. Equation 2 compresses the tactile field into one categorical decision before policy inference, whereas Equation 3 exposes spatially distributed tactile evidence to the policy objective.

Approach

Continuous Tactile-Visual Policy

The system observes RGB frames from two cameras and tactile frames from two gripper-mounted GelSight sensors, alongside proprioceptive joint state and a tokenized language prompt (Figure 2). Visual frames are encoded with SigLIP, as in the unmodified VLA backbone. Each GelSight sensor produces a 480×640 RGB frame, which is resized to 224×224 and independently encoded by a shared, pretrained DINOv2-Base (ViT-B/14) encoder (Oquab et al. 2024). The class token is discarded, leaving 256 patch tokens of dimension 768 from each sensor; concatenating the two feature sequences yields 512 tactile tokens. DINOv2's patch representation preserves local geometric structure from the deformation image rather than reducing the complete contact to a class score.

A jointly trained linear projection $W \in \mathbb{R}^{2048 \times 768}$ maps each tactile token to the Gemma 2B backbone dimension. The projected tactile tokens are inserted immediately after the SigLIP visual tokens and before the language tokens, following the ordering *visual, tactile, language, joint-state*. We add neither modality-specific attention blocks nor type embeddings; the backbone assigns ordinary sequential position IDs. The DINOv2 encoder, projection layer, and π_0 backbone are fully fine-tuned end to end with the same flow-matching objective. A fixed task prompt—"Pick up the avocado and place it into the corresponding basket"—is used throughout execution.

The policy predicts action chunks of 50 consecutive joint-position targets (six arm joints and one gripper position); at each chunk boundary, it encodes a fresh tactile pair with the current visual observation and then executes the resulting pre-committed targets without mid-chunk tactile correction. This cadence is sufficient for the sorting task because the ripeness decision is encoded during the grasp-contact chunk and sustained through the subsequent sorting chunk.

This design preserves the pretrained VLA and visual-encoder architectures without modification beyond the projection layer, avoiding the additional parameters and computational overhead of cross-modal attention while allowing the backbone to implicitly learn intermodal dependencies during fine-tuning.

Table 1 summarizes the model interface at one chunk boundary. The tactile pathway is the only newly introduced token source: all other inputs and the 50×7 action target retain the original policy structure. This separation is useful for implementation because the fusion model can reuse the existing observation preprocessing, language tokenizer, proprioceptive projection, and action decoder.

The 512-token tactile sequence is obtained without spatial pooling: each sensor contributes 256 DINOv2 patch tokens, and the shared linear layer projects every token independently. Consequently, the projection adds no cross-sensor mixing before the backbone; interaction between fingers and with RGB context occurs only inside the jointly fine-tuned policy. This choice retains local deformation and left–right asymmetry but lengthens the context substantially relative to a single class label, motivating the latency comparison in the Inference Cost section below.

Signal	Encoder	Policy representation
RGB (2)	SigLIP	Native visual tokens
Touch (2×224^2)	DINOv2-B/14	512 tokens, 2048D
Language	Tokenizer	Prompt tokens
State (7D)	MLP	Backbone-aligned state
Action	Flow head	50×7 targets

Table 1: Inputs and representations at one action-chunk boundary.

Comparison Policies

We compare three systems built on the same π_0 backbone. **Vanilla VLA** receives the two RGB views, joint state, and the fixed task prompt, but no tactile observation. **Parallel inference** runs a YOLO tactile classifier beside the VLA. The classifier selects among three prompts: "Pick up the avocado" for *empty*, "Put [it] into the right basket" for *ripe*, and "Put [it] into the left basket" for *firm*. The predicted prompt is supplied at the same chunk boundary at which continuous fusion receives its tactile frames. **Tactile-visual fusion** is the proposed model above, which retains a fixed prompt and supplies the projected DINOv2 patch sequence. Thus all three systems share RGB observations, proprioception, action representation, and backbone family, while the two tactile systems differ in several coupled design choices summarized later in Table 4.

Figure 3 makes the intervention point explicit. Vanilla VLA and parallel inference retain the unmodified policy input structure; the latter changes only the language content selected by an external classifier. The proposed system changes the policy context itself by adding tactile tokens. The classifier baseline tests whether a low-bandwidth categorical decision is sufficient when delivered through an already pretrained interface, whereas feature fusion tests whether retaining the deformation field within the learned policy context improves behavioral conditioning.

Experiments

Robot Platform and Data

Experiments use a 6-DOF TrossenAI robotic arm with its integrated 1-DOF gripper. Two Intel RealSense D405 cameras provide RGB observations: one is wrist-mounted and the other provides a fixed side view of the workspace. Two GelSight sensors, chosen for their high-resolution imaging and ease of gripper mounting, are attached to the gripper fingers using 3D-printed mounts. The recorded proprioceptive state comprises the six arm-joint angles and gripper position.

Fifty teleoperated sorting demonstrations are collected with 10 training avocados spanning two ripeness classes (ripe and firm). Samples are placed at varied orientations and positions across the reachable workspace. Every trajectory records synchronized RGB and tactile frames and the seven-dimensional robot state. For tactile-classifier training, frames are labeled as *ripe*, *firm*, or *empty*; the last label represents the pre-contact portion of the trajectory. Evaluation

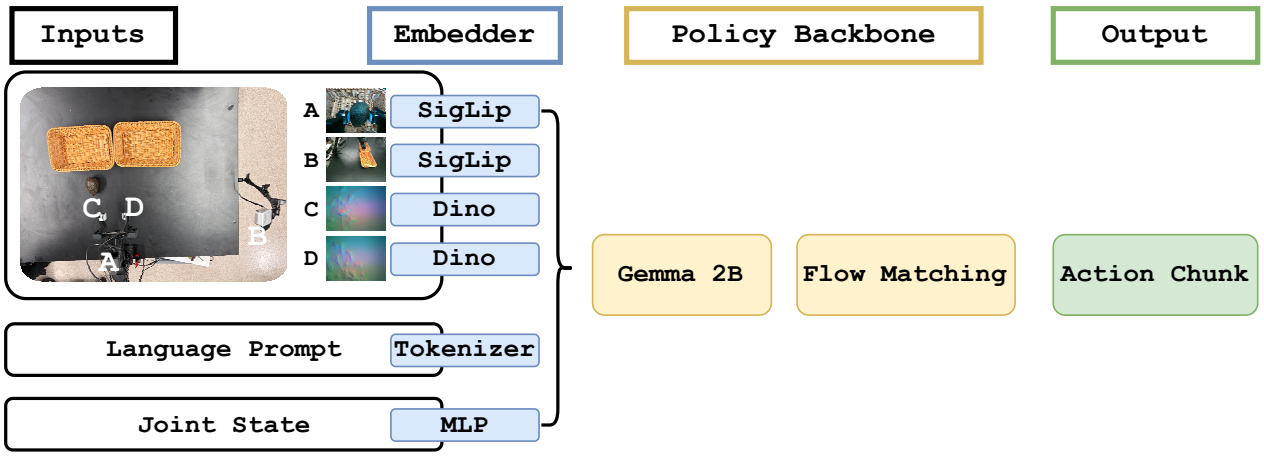


Figure 2: Overview of the proposed architecture. RGB frames (A, B) are encoded via SigLIP; frames from the two gripper-mounted tactile sensors (C, D) are independently encoded by a shared DINOv2 encoder, after which their feature sequences are concatenated. The resulting representations are combined with the tokenized language prompt and MLP-projected joint state, then passed to the π_0 (Gemma 2B) policy backbone with flow matching to generate action chunks.

uses a separate pool of 10 avocados not used for demonstration collection.

Training and Implementation

The two VLA baseline configurations and the fusion model are fully fine-tuned for 30,000 steps with batch size 32 on an NVIDIA RTX Pro 6000 (96 GB), requiring approximately 25 hours per model. The YOLO tactile classifier is trained separately for up to 500 epochs with early-stopping patience of 100 on an NVIDIA RTX 4090 (24 GB). Frames from the left and right tactile sensors are concatenated vertically and resized to 640×640. A 75/25 train-validation split is used over the three contact classes.

Observation and Action-Chunk Alignment

Each query groups the two RGB views, two tactile views when applicable, seven-dimensional robot state, and prompt available at one chunk boundary, then predicts the next 50 joint-position targets. This preserves causal ordering: pre-contact queries contain unloaded tactile images, contact queries contain deformation evidence, and transport queries combine recent contact state with the evolving visual trajectory. The training target is always the subsequent action chunk, so future tactile frames are not exposed.

The tactile stream is retained throughout the trajectory instead of being cropped to hand-selected contacts. Unloaded frames serve as negative examples, while independently encoded left and right frames preserve asymmetric contact. The policy receives no contact flag, force threshold, or manually selected ripeness timestep; it must learn when touch becomes informative and how long it should affect later chunks.

Tactile Classifier Verification

The classifier is a diagnostic baseline rather than the main perception contribution. Figure 4 shows that all 165 validation frames are assigned correctly in the chosen split, although correlated frames from the same contact sequence limit the generalization implied by this result. We therefore also check predictions during robot evaluation. The classifier supplies the correct category in all 20 parallel-inference trials, including the four sorting failures. These failures occur downstream of the observed categorical decision, supporting only the narrow conclusion that the complete classifier-to-prompt interface is less reliable than the tested fusion interface under this protocol.

Evaluation Protocol and Metrics

All policies are tested on the same 10 held-out avocados (5 ripe, 5 firm). The workspace contains a fixed firm basket on the left and ripe basket on the right. We define five start positions distributed across the reachable region. One specimen from each class is assigned to each position and tested twice under each method, yielding 20 trials per method. Positions near the opposite-class basket require the robot to bypass that basket before reaching the correct destination, reducing the usefulness of a simple positional shortcut. A trial succeeds only if the robot grasps the avocado and releases it into the bin matching its ripeness class. Because each specimen is observed twice, the 20 trials are repeated measurements on 10 biological samples rather than 20 independent specimens.

We report sorting success and wall-clock policy latency from sensor input to the next action chunk. Latencies are averaged over 20 inference calls. For a descriptive uncertainty estimate matching the prior analysis, we report Wilson score intervals for k successes in n trials:

$$\frac{2n\hat{p} + z^2 \pm z\sqrt{z^2 + 4n\hat{p}(1-\hat{p})}}{2(n+z^2)}, \tag{4}$$

where $\hat{p} = k/n$ and $z = 1.96$ for 95% coverage (Wilson 1927). These intervals treat trials as Bernoulli observations

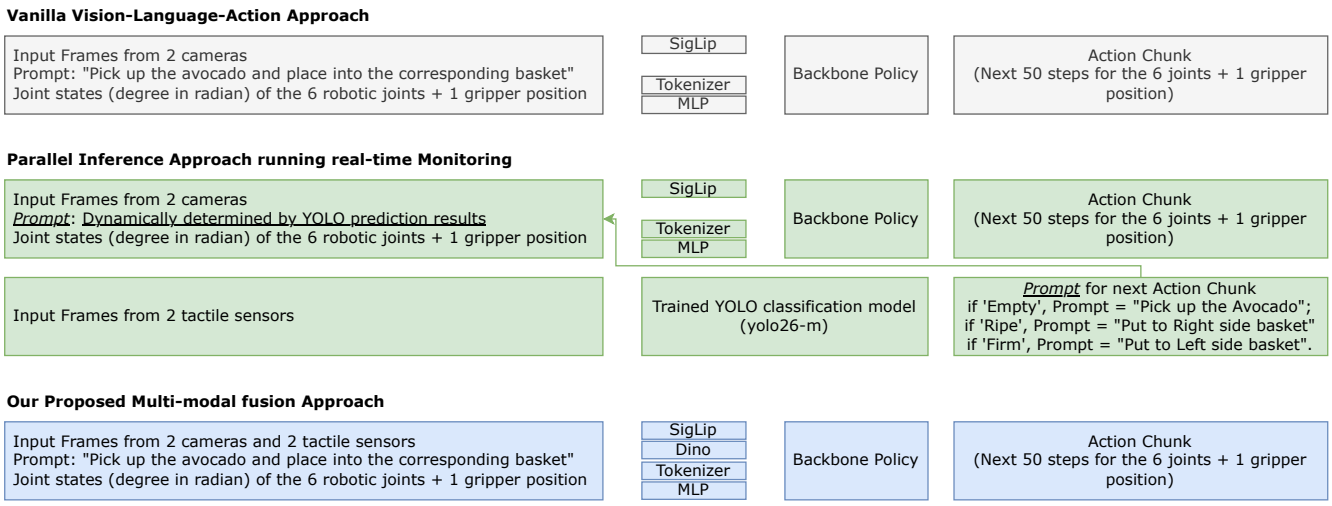


Figure 3: Information flow in the three evaluated systems. Top: vanilla VLA observes RGB, language, and robot state. Middle: parallel inference converts the tactile classifier output into a dynamic prompt. Bottom: the proposed policy supplies projected tactile features directly to the backbone while retaining a fixed prompt.

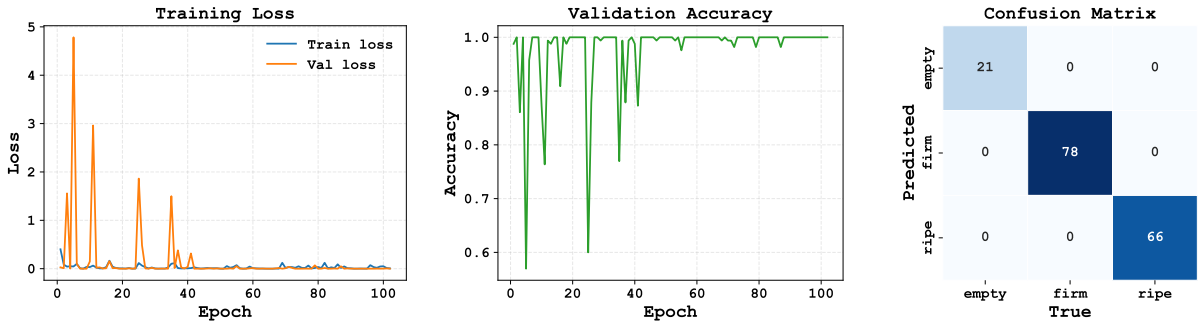


Figure 4: Training and validation loss, validation top-1 accuracy, and the frame-level validation confusion matrix for the auxiliary tactile classifier. The matrix contains 21 empty, 78 firm, and 66 ripe validation frames, with no observed off-diagonal errors.

Method	Correct	Success	Wilson 95% CI
Vanilla VLA	14/20	70%	[48.1, 85.5]
Parallel Inference	16/20	80%	[58.4, 91.9]
Tactile-Visual (Ours)	19/20	95%	[76.4, 99.1]

Table 2: Sorting success over 20 trials per method on the same 10 held-out avocados. Intervals are percentages and are descriptive because trials repeat specimens.

and therefore do not model within-specimen dependence; we use them as descriptive uncertainty, not as evidence of statistical significance.

Results

Sorting Performance

Table 2 shows that the proposed policy completes 19 of 20 trials, compared with 16 for parallel inference and 14 for vanilla VLA. The 25-point difference from vanilla VLA supports the practical value of tactile input for a visually ambiguous sorting criterion, while the additional 15-point difference from parallel inference indicates greater downstream reliability for feature fusion than classifier-to-prompt conditioning. The Wilson intervals overlap substantially, so the study does not establish a statistically significant population-level advantage.

Because the same specimens recur across methods, we also compare outcomes per specimen rather than per trial. Across the 10 held-out avocados, fusion never sorts a specimen worse than vanilla VLA, tying on 6 and improving on 4, with zero reversals, while the number of specimens where parallel inference and vanilla VLA disagree is too small (4) to support a separate test. A sign test on the 4 non-tied specimens ($p \approx 0.13$) does not reach significance at this sample size, but the direction is fully consistent with the aggregate result above.

Table 3 localizes the failures. Vanilla VLA failures concentrate in firm samples, especially at S4 and S5, and

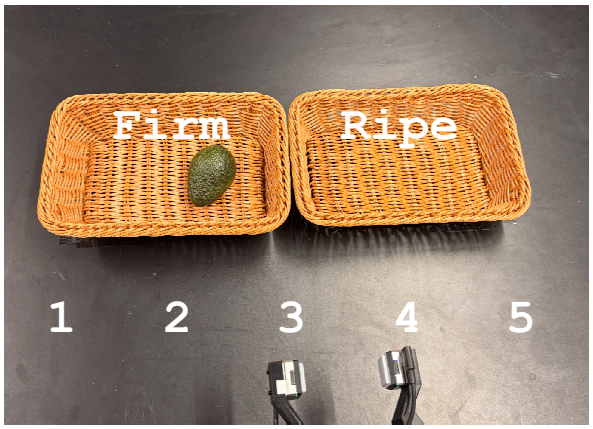


Figure 5: Evaluation workspace with fixed Firm and Ripe baskets and five test positions. Positions near the opposite-class basket require the policy to pass or bypass the incorrect destination.

Method	Class	S1	S2	S3	S4	S5
Vanilla	Firm	2/2	2/2	2/2	0/2	0/2
VLA	Ripe	1/2	1/2	2/2	2/2	2/2
Parallel	Firm	1/2	2/2	2/2	1/2	0/2
inference	Ripe	2/2	2/2	2/2	2/2	2/2
Tactile-	Firm	2/2	2/2	2/2	1/2	2/2
visual	Ripe	2/2	2/2	2/2	2/2	2/2

Table 3: Correct sorting trials at five start positions. Each cell reports successes out of two repeated trials for the assigned specimen.

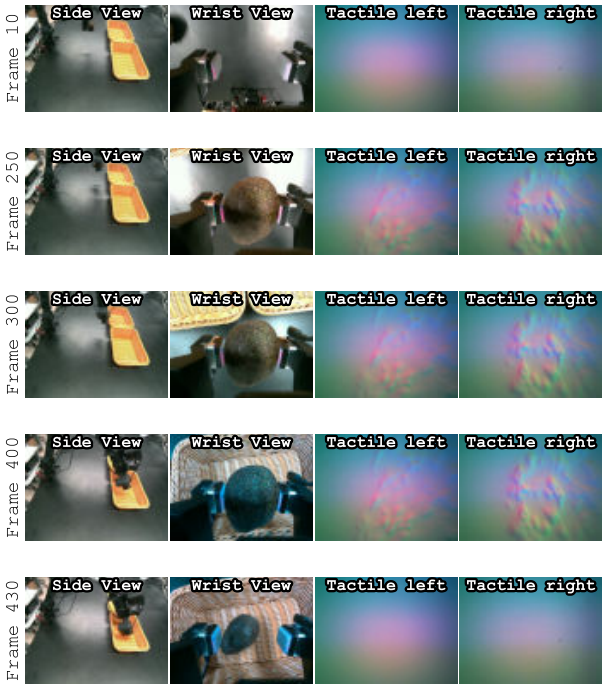


Figure 6: Representative tactile-visual rollout for a ripe avocado. Before contact (Frame 10), tactile images remain near their unloaded appearance. Contact produces structured deformation in Frames 250–400; the robot then transports the fruit to the Ripe basket and the sensors return toward baseline after release (Frame 430).

the parallel pipeline retains firm-class failures. Both tactile pipelines sort every ripe trial correctly. Feature fusion reduces the firm failures from four in parallel inference to one, with the remaining error at S4. This location lies close to the incorrect basket and yields a trajectory that passes near it; an atypical grasp region may also produce deformation patterns underrepresented in training. These are plausible mechanisms rather than separately validated causal explanations.

Class-Wise Pattern

The differences are driven primarily by firm avocados: both tactile pipelines sort every ripe trial correctly, while fusion recovers three of the four firm-class failures left uncorrected by parallel inference. This is a failure-localization result, not evidence that firm fruit is universally harder: lighting, skin appearance, binary labels, and only ten repeated trials per class limit generalization.

Behavior Across the Manipulation Sequence

The tactile observations are not equally informative throughout a rollout. Before contact, both GelSight frames show their unloaded appearance and the appropriate behavior is primarily vision-guided reaching. During gripper closure, the fruit surface deforms the elastomer, producing the spatial patterns illustrated in Figure 1. Ripeness becomes observable only in this contact phase. After lift-off, the policy must preserve the inferred distinction while executing a destination-dependent transport trajectory. This temporal structure makes the task different from either static tactile classification or short-horizon reactive force control: useful evidence arrives after the action sequence has begun, and its effect must persist after the most discriminative contact frames have passed.

The perfect grasp rate across the evaluated trials provides an additional diagnostic separation. Failures are not caused by inability to locate or pick up the fruit; they occur when a successfully grasped avocado is carried to the wrong basket. The outcome being changed by tactile integration is therefore consistent with ripeness-conditioned destination selection rather than a generic improvement in grasp execution. This does not prove how the backbone represents ripeness internally, but it narrows the observed behavioral effect to the phase of the task in which the latent property should matter.

Signal-Pathway Comparison

Table 4 characterizes the coupled differences among the complete pipelines. The discrete tactile pathway gains 10 points over vanilla VLA, and fusion adds another 15 points. Because the latter comparison jointly changes encoder, representation, injection point, prompt strategy, and optimiza-

Variant	Encoder	Fusion	Prompt	E2E
Vanilla VLA	None	None	Fixed	–
Parallel Inf.	YOLO	Text tok.	Dynamic	No
Ours	DINOv2	Patch tok.	Fixed	Yes

Table 4: Differences among the three pipelines in encoder, representation and injection pathway, prompt strategy, and end-to-end training.

Component or policy	Latency (ms)	Device
YOLO tactile classifier	8.7	RTX 5060
Vanilla π_0	66.2	RTX 4090
Tactile-visual π_0	71.3	RTX 4090

Table 5: Mean latency over 20 inference calls. YOLO runs concurrently with the VLA on a separate GPU in the parallel system.

tion, it is a system comparison rather than evidence for any single factor; the Interpreting the Fusion Gap section below examines the plausible mechanisms.

Inference Cost

As shown in Table 5, continuous fusion adds 5.1ms, or 7.7%, over the vision-only VLA on the same RTX 4090. The parallel system's effective critical-path latency is bounded by the 66.2ms VLA call because its 8.7ms classifier runs concurrently, but it uses a second GPU. This comparison therefore exposes a resource tradeoff hidden by wall-clock latency: the proposed model is slightly slower on one policy GPU, whereas parallel inference maintains the vanilla critical path by adding a classifier device. The classifier runs on a separate device so its execution proceeds in parallel with, and is not blocked by, the VLA's chunk execution, though in this evaluation its prediction is still sampled once per chunk boundary for a controlled comparison.

Discussion

Interpreting the Fusion Gap

A concrete hypothesis for the 15-point gap is representational bandwidth. Numerically, fusion supplies 512 tactile tokens at the 2048-dimensional backbone width ($1{,}048{,}576$ scalar entries) at each chunk boundary, whereas the classifier alters only 2–3 language tokens among its three prompts ($4{,}096$–$6{,}144$ entries at the same width), a roughly 171–$256\times$ difference in raw input footprint rather than an information-theoretic measure. Parallel inference vertically concatenates the two tactile frames, compresses them to one of three categories, and communicates the result through the few language tokens that distinguish the prompts. Fusion independently encodes both frames and retains their patch-token sequences, preserving spatial variation between sensors. This richer input may condition behavior more strongly, but representation size does not prove causal use of touch. Fusion location, encoder, and optimization also differ: YOLO is trained separately and enters through language, whereas projected DINOv2 features are jointly fine-tuned beside visual tokens.

There are at least four non-exclusive explanations for why a correct class prompt may fail to control the action. First, a categorical prompt removes within-class information about contact strength, asymmetry between fingers, and spatial deformation. Those features may help the policy distinguish a confident contact from a marginal or atypical grasp. Second, the prompt enters through a pretrained language pathway whose semantics are learned mostly from vision-language data rather than from paired tactile-action experience. Fine-tuning can associate the words *firm* and *ripe* with destinations, but visual priors and trajectory context may still dominate this small token difference. Third, the fused encoder and policy are optimized jointly, allowing the tactile representation to adapt directly to the action objective, whereas the classifier is optimized to predict labels independently. Fourth, the parallel pipeline's prompts differ not only in bandwidth but in linguistic form: distinguishing "right basket" from "left basket" requires the backbone to resolve a spatial relation from a short phrase, a class of instruction on which vision-language models are known to be comparatively unreliable. Part of the observed gap may therefore reflect general spatial-language grounding weakness rather than tactile bandwidth specifically; disentangling the two would require comparing spatial prompts against equivalent non-spatial prompts (e.g., a color- or name-based basket reference) under the same classifier.

Alternative Designs and Decisive Ablations

Several intermediate systems would isolate the factors above. To isolate representation granularity while holding the encoder fixed, DINOv2 patch tokens could be pooled and classified, with the resulting label injected either as text or as a learned class embedding. Conversely, the continuous DINOv2 representation could be injected through a dedicated prompt adapter while keeping its dimensionality approximately fixed. To isolate end-to-end optimization, the DINOv2 encoder and projection could be frozen, then compared with joint fine-tuning. Modality dropout during training would test whether the policy has learned redundant visual shortcuts: performance with tactile tokens masked at evaluation should approach the vision-only baseline if touch is causally used. Finally, tactile-patch occlusion, gradient saliency, or attention rollout could test whether contacts with larger deformation receive greater policy influence. The classifier's dynamic prompt could also be swapped between spatial (e.g., "right basket") and non-spatial (e.g., color- or name-based) basket references to test whether classifier-to-prompt failures reflect spatial-language grounding rather than category bandwidth.

The most direct next experiment is a matched DINOv2 class-versus-patch comparison, followed by frozen-versus-fine-tuned encoders. Mechanistic visualization should follow these behavioral controls because an attention map alone does not establish causal use of a modality.

Relevance to Representation Design

Beyond the specific fusion mechanism, the comparison suggests a candidate representation requirement for non-visual physical properties in policy learning: the representation

should preserve graded, spatially structured evidence and provide sufficient bandwidth at the policy input rather than compressing the observation to a categorical cue carried by a few prompt tokens. The fusion mechanism used here, token concatenation rather than added cross-modal attention, was a deliberate design choice. Cross-modal attention mechanisms typically introduce new learned parameters and a bespoke architecture per modality pairing, complicating reuse across different VLA backbones and sensor combinations. Our approach instead projects the tactile patch features into the backbone's shared token dimension and treats them as additional tokens in its existing sequence. Testing whether this pattern generalizes across backbones beyond π_0 and modalities beyond tactile sensing is a natural next step; if it does, it would suggest a reusable recipe for adding continuous physical-property representations to VLA manipulation pipelines without per-modality cross-attention design.

Application Scope and Deployment Trade-offs

A fixed automation cell may be preferable when produce arrives at a known pose on a conveyor, the decision boundary is stable, and throughput dominates reconfigurability. A dedicated firmness tester and mechanical diverter could then be cheaper, easier to certify, and faster than a general manipulation policy. Our experiment does not compare against such a cell and should not be read as evidence that a VLA is necessary for every avocado-sorting line. The VLA formulation is most relevant to variable-pose, small-batch, or shared-workspace settings in which the same robot must localize, grasp, assess, and route individual items without purpose-built fixturing. In that regime, integrating tactile quality evidence into an already capable manipulation policy avoids a separate classifier-control interface and may support reuse across related handling tasks. Establishing that systems-level advantage would require a future comparison of throughput, cost, damage rate, and reconfiguration effort against fixed automation.

Limitations and Future Work

This is a small-scale, early-stage result. Evaluation uses 10 held-out avocados disjoint from the 10 used for demonstrations, but remains limited to 20 physical specimens, two ripeness classes, one produce type, and one controlled setting. Each held-out specimen contributes two trials per method, so $n = 20$ contains repeated observations rather than 20 independent specimens, and confidence intervals overlap. We therefore do not claim statistical significance or deployment readiness. A per-specimen paired comparison (Sorting Performance) supports the same direction as the aggregate result but likewise cannot reach significance at this sample size. We also did not test every held-out specimen at every start position: doing so would require repeatedly regrasping the same avocado, and repeated contact risks bruising or otherwise altering its surface compliance between trials, confounding the tactile signal being measured. Separating positional from specimen-specific failure modes would instead require a larger pool of fresh specimens assigned one-to-one with positions, which our sample size did not support. Beyond the controlled ablations above, modality dropout and tactile-patch attribution could test causal use of touch, while broader validation must cover additional produce, ripeness stages, sensors, and VLA backbones.

Ripeness labels also require tighter operational definition. Future data should pair touch with an independent firmness measurement and record cultivar and storage conditions, enabling cross-day and cross-cultivar tests that distinguish compliance from batch-specific correlations.

Conclusion

We present a ripeness-conditioned tactile-visual policy built on π_0. It achieves 95% success versus 70% for vision only and 80% for classifier-to-prompt conditioning, with 7.7% inference overhead. The result favors the tested feature-fusion system but does not establish that token concatenation is universally optimal or isolate its coupled design factors. The finding is not specific to avocados: any contact-rich task where a policy must act on a physical property that is only tactile-observable, such as assembly tolerance, cable-routing tension, or fabric compliance, faces the same choice between a discrete classifier bottleneck and a continuous fused representation. The central observation is narrower: even a correct tactile category can fail to control a pretrained action policy when communicated only through a prompt.

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